



POLITECNICO
MILANO 1863

Apache Spark

Alessandro Margara

`alessandro.margara@polimi.it`

`https://margara.faculty.polimi.it`

Rules

- Rename the SparkGroupXX.java file replacing XX with the number of your group
- Write in the comment on top of the class your group number and the name of all group members
- Submit only a single java file with your solution
 - Submitted from the contact email provided in the group registration document

Assumptions

Four input datasets

1. profs

- Type: static, csv file
- Fields: prof_name, course_name

2. courses

- Type: static, csv file
- Fields: course_name, course_hours, course_students

3. videos

- Type: static, csv file
- Fields: video_id, video_duration, course_name

4. visualizations

- Type: dynamic, stream
- Fields: timestamp, value
- Each entry with value v indicates that someone watched video with video_id equal to v

Requirements

- For all queries: limit unnecessary recomputations as much as possible!
- For streaming queries: write the results on the console, showing only the results that changed since the last evaluation

Requirements

- Q1: compute the total number of hours of lecture for each prof
- Q2: for each course, compute the total number of visualizations of videos of that course, computed over 1 minute, updated every 10 seconds
- Q3: for each video, compute the total number of visualizations of that video with respect to the number of students in the course in which the video is used